

Worksheet 2

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1. Aim/Overview of the practical:

Interface a led, pushbutton (**Pull Up**) and LCD with Arduino Uno. Use two led and two pushbuttons. Use pushbutton as toggle switch.

- Case 1: When press button 1 once then LED 1 will turn ON and print “**Button1 ON**” in Row 1 and “**LED 1 ON**” in Row 2.
- Case 2: When we press button 1 again then LED 1 will turn OFF and print “**Button1 OFF**” in Row 1 and “**LED 1 OFF**” in Row 2.
- Case 3: When we press button 2 once then LED 2 will turn ON and print “**Button2 ON**” in Row 1 and “**LED 2 ON**” in Row 2.
- Case 4: When we release button 2 again then LED 2 will turn OFF and print “**Button2 OFF**” in Row 1 and “**LED2 OFF**” in Row 2.

2. Apparatus

Hardware Requirements: Arduino Uno R3 Board

LCD

LED

Pushbutton

Software requirements: Microsoft Windows 10

TinkerCAD

3. Circuit Diagram(TinkerCad):

Case 1:

Coding:

```
#include <LiquidCrystal.h>
```

```
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
```

```
const int button1 = 7;
```

```
const int button2 = 8;
```

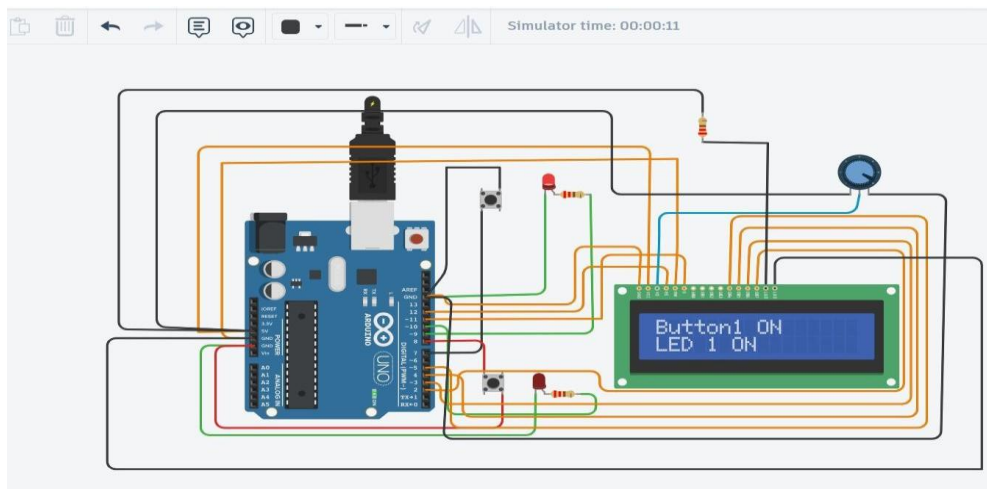
```
const int led1 = 9;
```

```
const int led2 = 10;
```

```
bool led1State = false;
```

```
int lastButton1 = HIGH;  
int lastButton2 = HIGH;  
  
void setup()  
{  
  pinMode(button1, INPUT_PULLUP);  
  pinMode(button2, INPUT_PULLUP);  
  
  pinMode(led1, OUTPUT);  
  pinMode(led2, OUTPUT);  
  
  lcd.begin(16, 2);  
  lcd.clear();  
  
  lcd.setCursor(0, 0);  
  lcd.print("Ready");  
  
  digitalWrite(led1, LOW);  
  digitalWrite(led2, LOW);  
}
```

Output:



Case 2:

Coding:

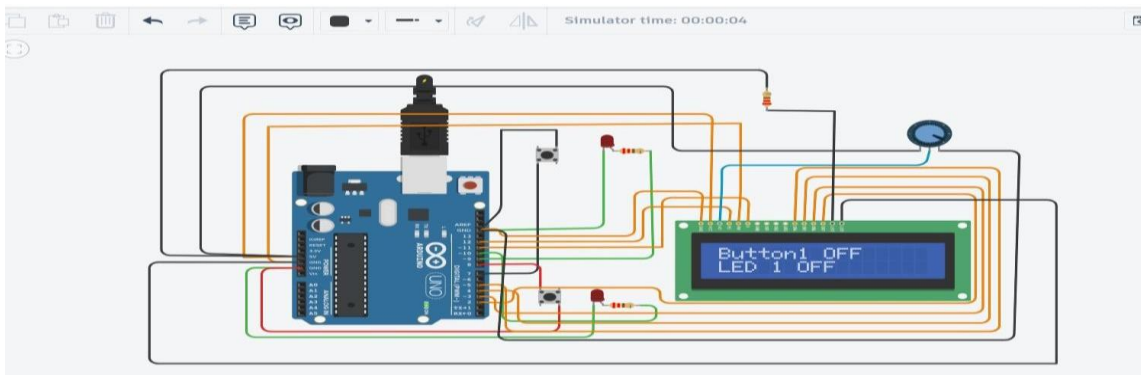
```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
```

```
const int button1 = 7;
const int button2 = 8;
const int led1 = 9;
const int led2 = 10;
```

```
bool led1State = false;
```

```
int lastButton1 = HIGH;
int lastButton2 =
HIGH;.....
.....
```

Output:



Case 3:

Coding:

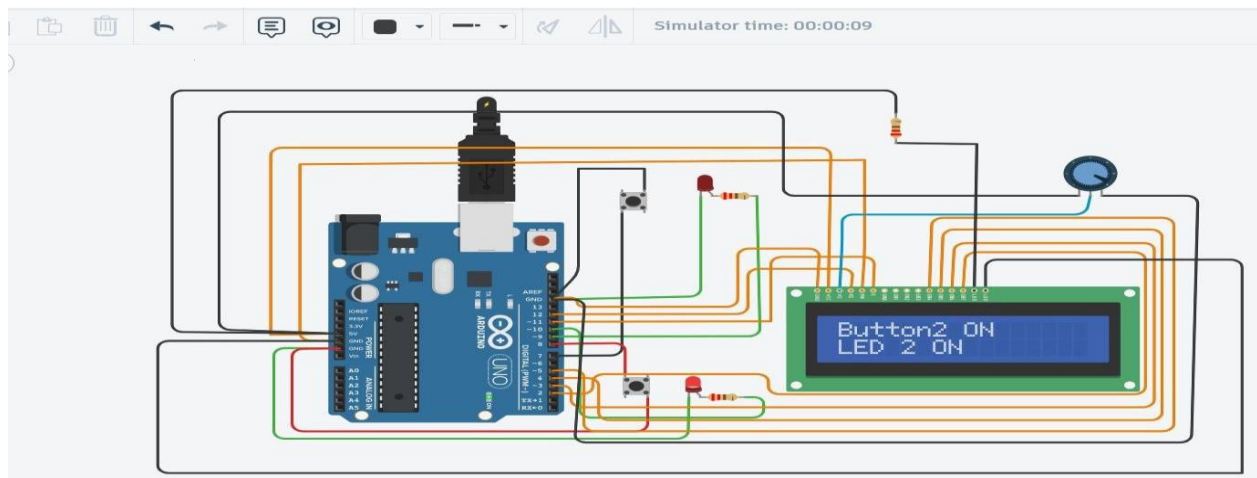
```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
```

```
const int button1 = 7;
const int button2 = 8;
const int led1 = 9;
const int led2 = 10;
```

```
bool led1State = false;
```

```
int lastButton1 = HIGH;
int lastButton2 = HIGH;
void setup()
{
  pinMode(button1, INPUT_PULLUP);
  pinMode (button2, INPUT_PULLUP);
  pinMode (led1, OUTPUT);
  pinMode (led2, OUTPUT);
}
```

Output:



Case 4:

Coding:

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
const int button1 = 7;
const int button2 = 8;
const int led1 = 9;
const int led2 = 10;
```

```
bool led1State = false;
```

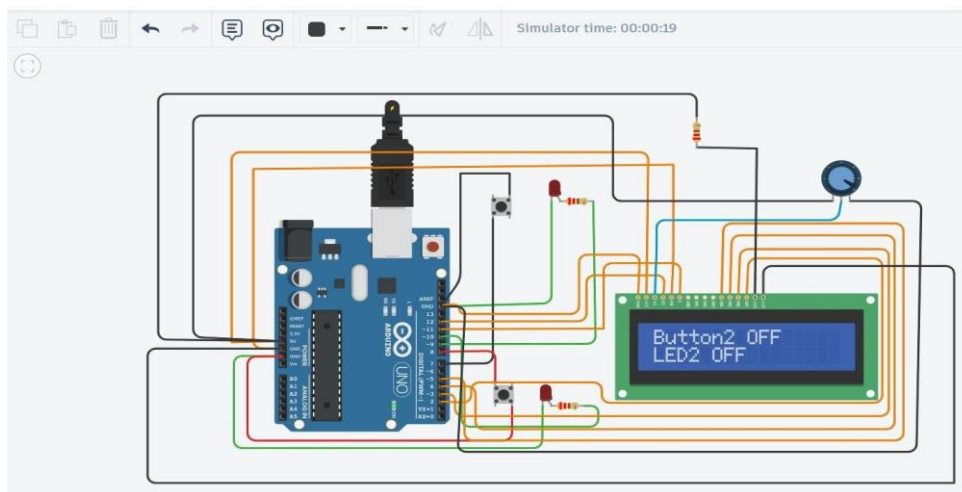
```
const int led2 = 10;

bool led1State = false;

int lastButton1 = HIGH;
int lastButton2 = HIGH;
void setup()
{
  pinMode(button1, INPUT_PULLUP);
  pinMode(button2, INPUT_PULLUP);
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);

  lcd.begin(16, 2);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Ready")
}
```

Output:



5. Learning outcomes (What I have learnt):

1. Using a 16*2 LCD in circuit.
2. Circuit formation using Arduino uno R3, push button, LED, LCD and potentiometer.
3. Integrating a potentiometer in a circuit.
4. Concepts of pull up and pull down in push button.
5. Components of 16*2 LCD

